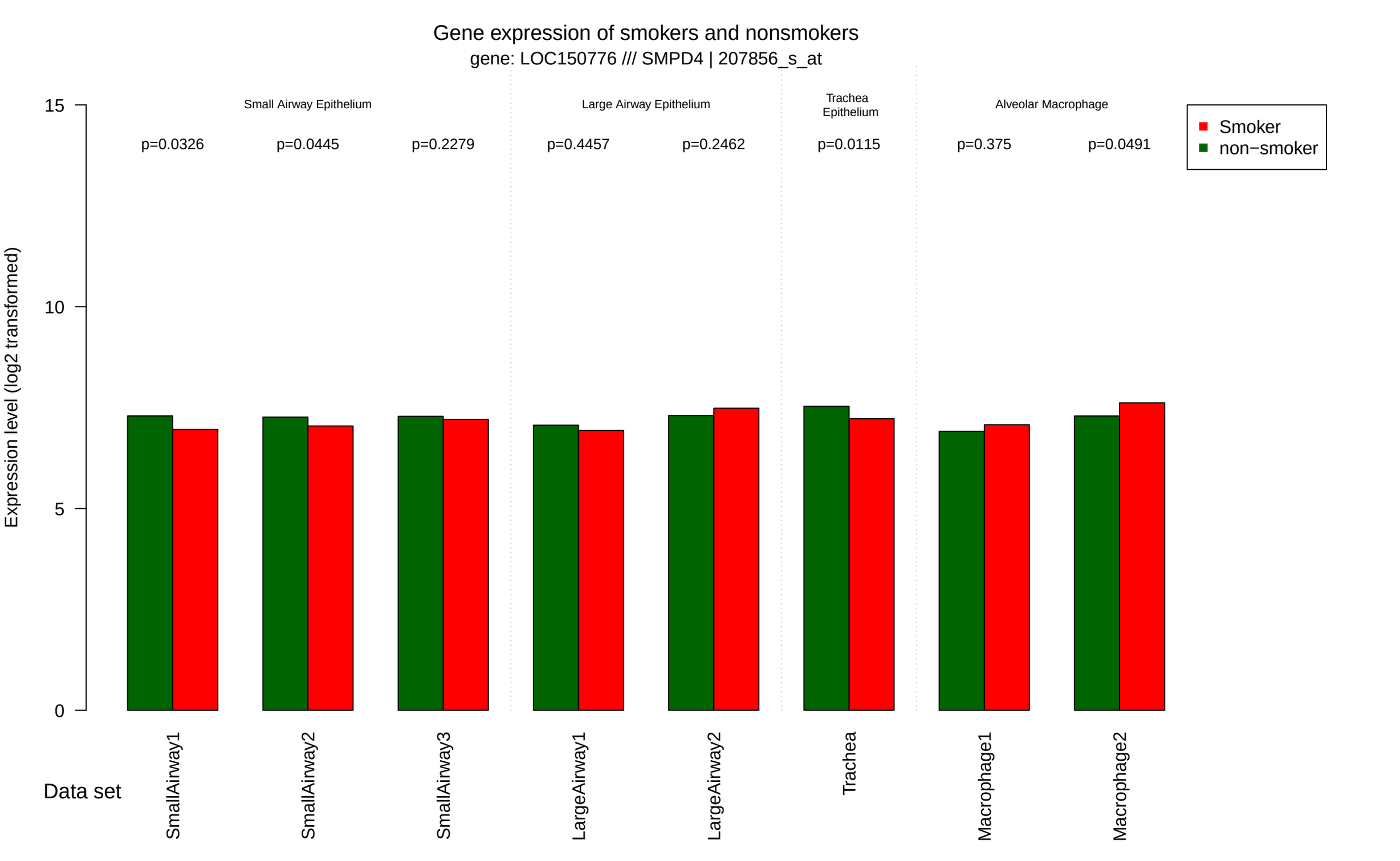
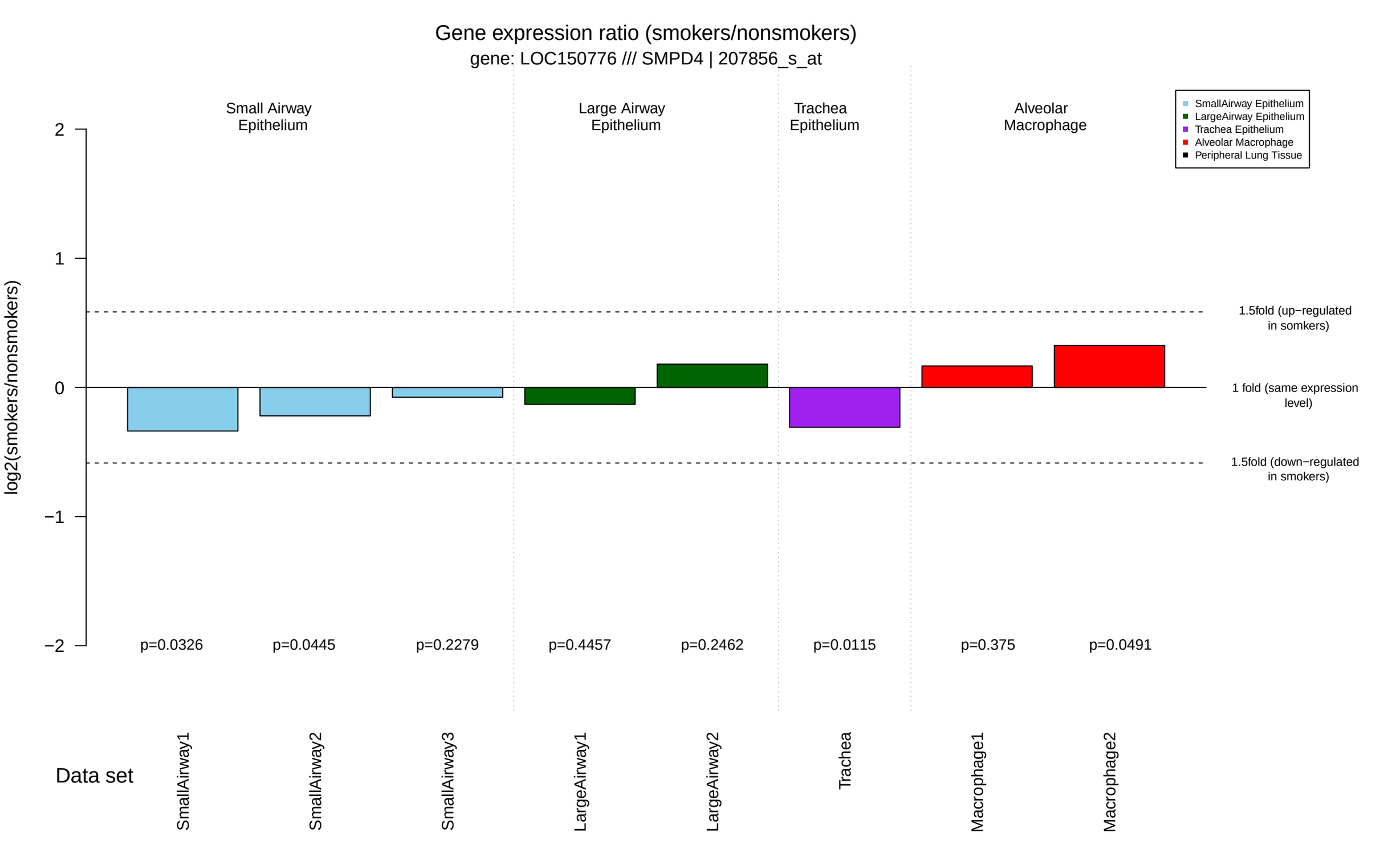
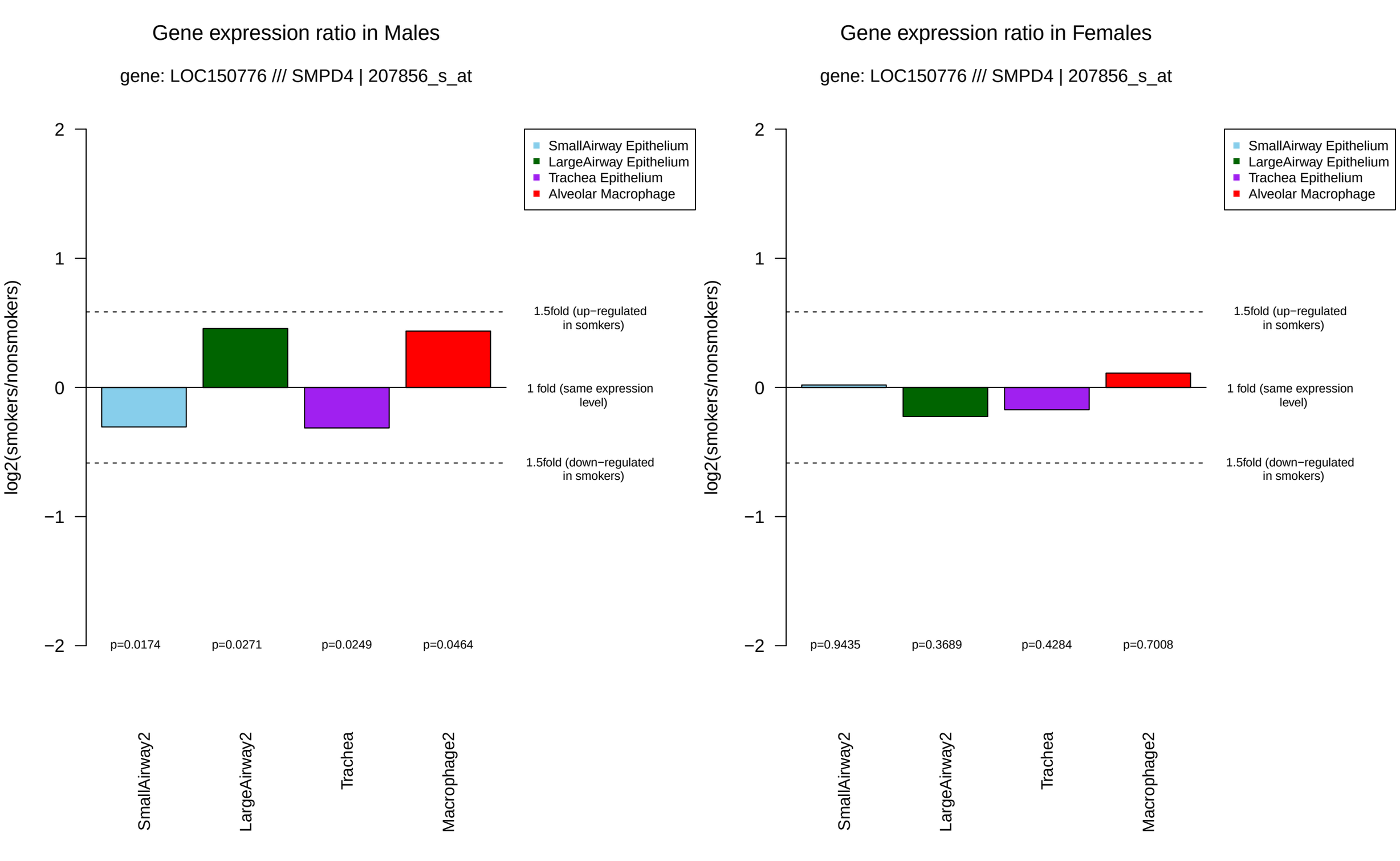
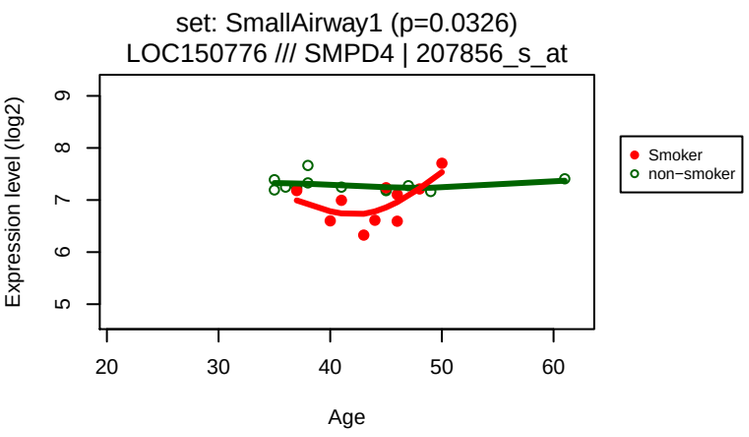
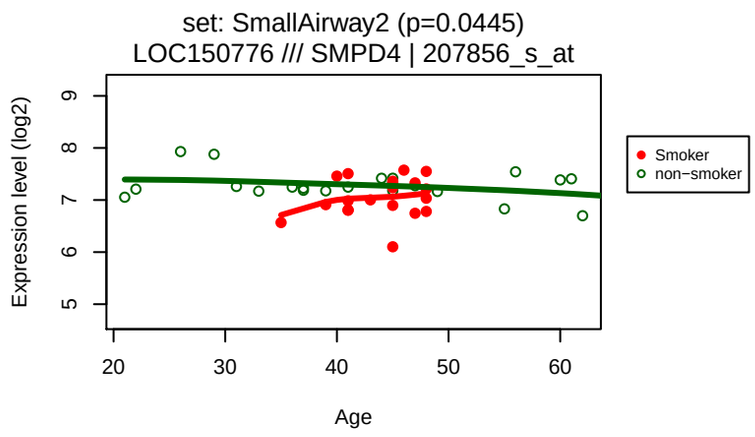
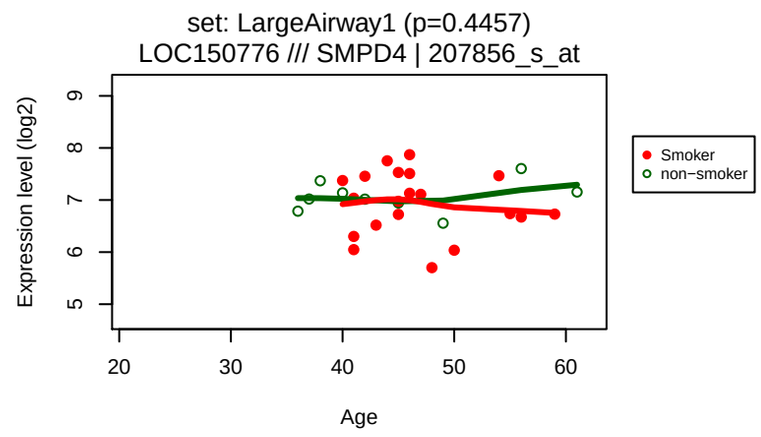

Gene expression of smokers and nonsmokers
gene: LOC150776 /// SMPD4 | 207856_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0326
p=0.0445
p=0.2279
p=0.4457
p=0.2462
p=0.0115
p=0.375
p=0.0491
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

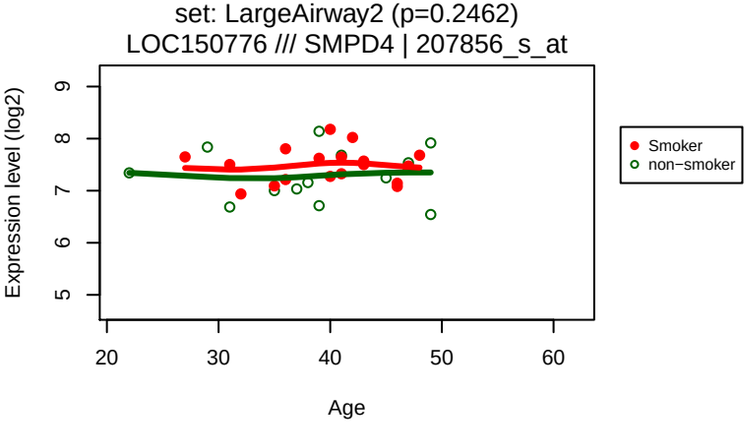

Gene expression ratio (smokers/nonsmokers)
gene: LOC150776 /// SMPD4 | 207856_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0326
p=0.0445
p=0.2279
p=0.4457
p=0.2462
p=0.0115
p=0.375
p=0.0491
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

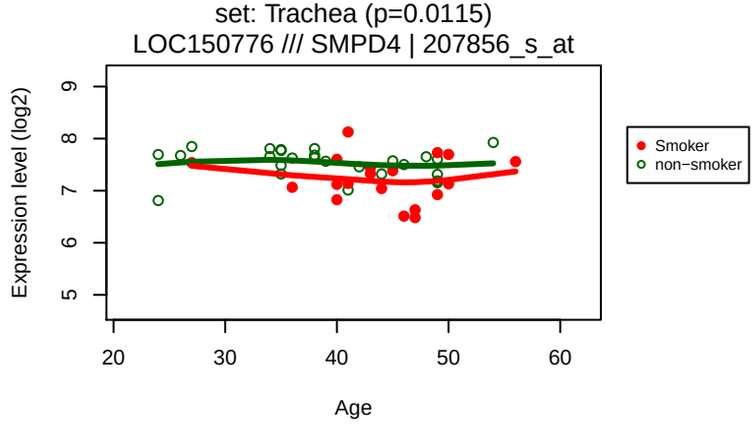

Gene expression ratio in Males
gene: LOC150776 /// SMPD4 | 207856_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0174
p=0.0271
p=0.0249
p=0.0464
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: LOC150776 /// SMPD4 | 207856_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.9435
p=0.3689
p=0.4284
p=0.7008
SmallAirway2
LargeAirway2
Trachea
Macrophage2

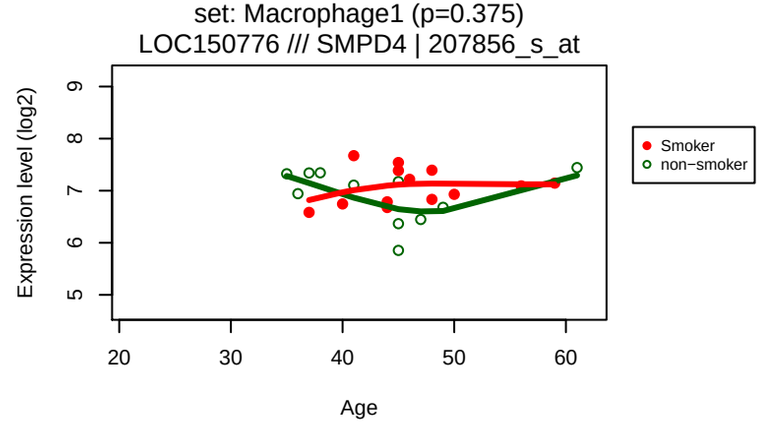

set: SmallAirway1 (p=0.0326)
LOC150776 /// SMPD4 | 207856_s_at
Expression level (log2)
Age
Smoker
non−smoker

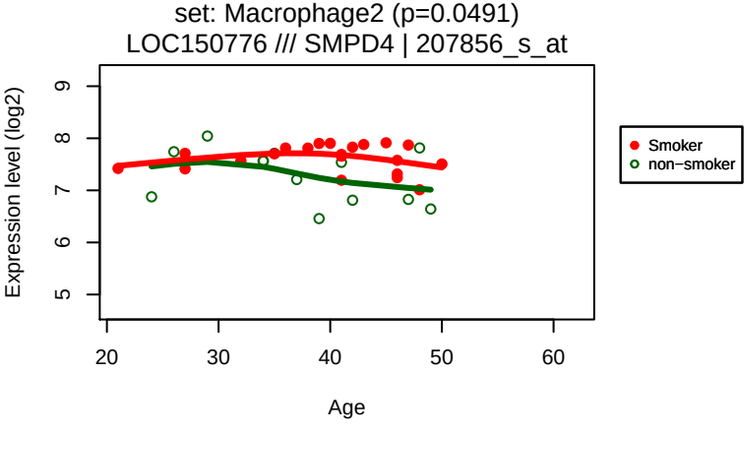

set: SmallAirway2 (p=0.0445)
LOC150776 /// SMPD4 | 207856_s_at
Expression level (log2)
Age
Smoker
non−smoker

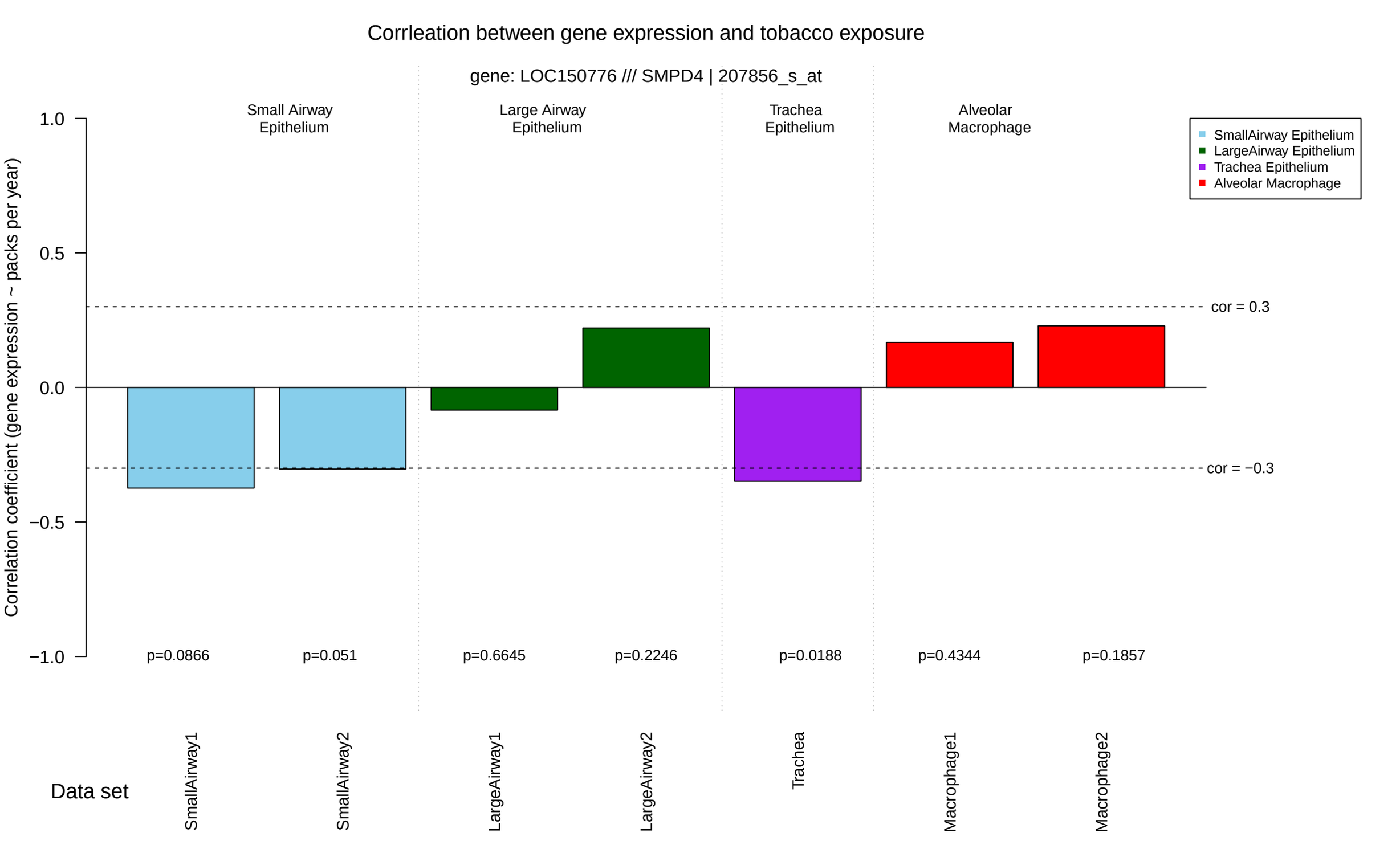

set: LargeAirway1 (p=0.4457)
LOC150776 /// SMPD4 | 207856_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.2462)
LOC150776 /// SMPD4 | 207856_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0115)
LOC150776 /// SMPD4 | 207856_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.375)
LOC150776 /// SMPD4 | 207856_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.0491)
LOC150776 /// SMPD4 | 207856_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: LOC150776 /// SMPD4 | 207856_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.0866
p=0.051
p=0.6645
p=0.2246
p=0.0188
p=0.4344
p=0.1857
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2